INCH-POUND

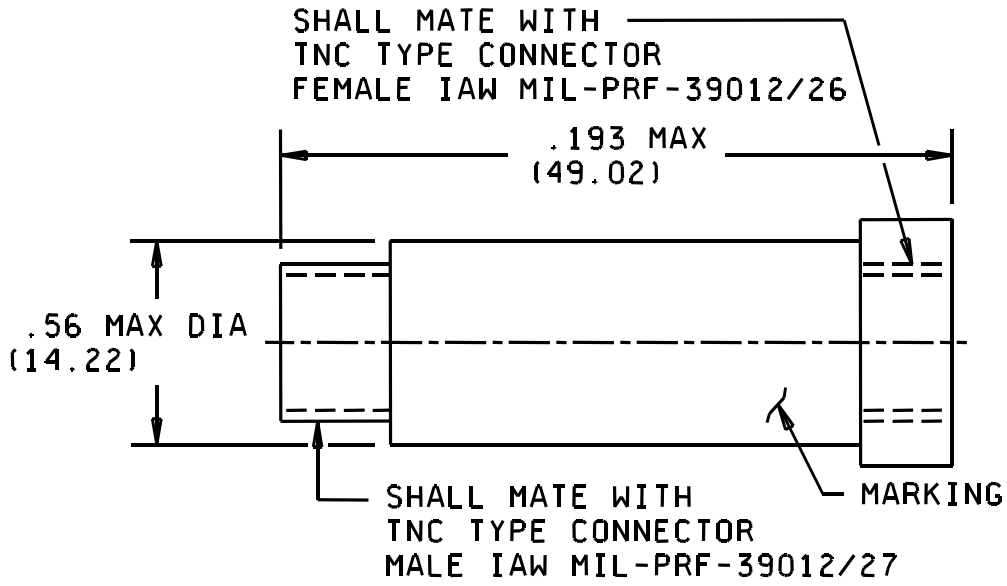
MIL-DTL-3933/17B
7 November 2002
SUPERSEDING
MIL-A-3933/17A
24 April 1985

DETAIL SPECIFICATION SHEET

ATTENUATORS, FIXED, COAXIAL LINE (TNC CONNECTORS), FREQUENCY RANGE: DC TO 18 GHz, LOW POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and the latest issue of MIL-DTL-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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ENGINEERING DATA:

Attenuators for use in equipment in which precision and stability are a secondary consideration.

Operating frequency range: DC - 4.5 GHz for PINs M3933/17-01 through -06; DC - 18 GHz for PINs M3933/17-07 through -24.

Test frequency ranges for SWR and attenuation measurements, measured or recorded, when using an Automatic Network Analyzer (ANA) or equivalent:

100 MHz (or lower) to 4.5 GHz for PINs M3933/17-01 through -06;
100 MHz (or lower) to 18 GHz for PINs M3933/17-07 through -24.

Basic test frequencies for sensitive measurements like connector repeatability, temperature sensitivity, power sensitivity and any other measurements where a 'delta' spec is imposed:

2.0 GHz for PINs M3933/17-01 through -06;
8.0 GHz for PINs M3933/17-07 through -24.

Operating temperature range: -55°C to 125°C.

Maximum weight: See table I.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Nominal impedance: 50 ohms.

Attenuation: See table I. Stability and sensitivity:

Maximum change in attenuation after:	up to 10 dB, inclusive (dB)	over 10 dB (dB/dB)
temperature change or thermal shock	0.5	0.05
vibration or shock	0.5	0.05
moisture resistance or salt spray	0.5	0.05
peak power	0.5	0.05

Frequency sensitivity, maximum: 0.4 dB/dB/GHz.
Temperature sensitivity of attenuation, maximum: 0.0006 dB/dB/°C.

Power: See table I.
Power sensitivity for full input power, maximum: 0.005 dB/dB/Watt.

VSWR: See table I.

Connector repeatability, maximum variation in attenuation: 0.04 dB.

Part or Identifying Number (PIN): M3933/17- (and dash number from table I).

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TABLE I. Electrical characteristics.

Dash number		Attenuation dB				Maximum power input at 25°C		VSWR maximum 1/					Weight (oz) Max	Dimension A (inches) maximum
		Nominal		Deviation										
				DC to 4.5 GHz	DC to 18 GHz	AV (W) 3/ (continuously)	Peak 2/ (kW)	DC to 4.5 GHz	DC to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
01	NS	1		±0.3	---	4	0.2	1.25:1	---	---	---	---	3.0	---
02	NS	2		±0.4	---	4	0.2	1.25:1	---	---	---	---	3.0	---
03	NS	3		±0.4	---	4	0.2	1.25:1	---	---	---	---	3.0	---
04	NS	4		±0.4	---	4	0.2	1.25:1	---	---	---	---	3.0	---
05	NS	5		±0.4	---	4	0.2	1.25:1	---	---	---	---	3.0	---
06	NS	6		±0.4	---	4	0.2	1.25:1	---	---	---	---	3.0	---
07	NS	20		---	±0.6	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
08	NS	30		---	±0.9	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
09	NS	1		---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
10	NS	2		---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
11	NS	3		---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
12	NS	4		---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
13	NS	5		---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash number		Attenuation dB				Maximum power input at 25°C		VSWR maximum 1/					Weight (oz) Max	Dimension A (inches) maximum
		Nominal	Deviation			AV (W) 3/ (continuously)	Peak 2/ (kW)	DC to 4.5 GHz	DC to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
				DC to 4.5 GHz	DC to 18 GHz									
14	N S	6		---	±0.4	2	0.2	---	1.15:1	1.20:1	1.15:1	1.35:1	3.0	---
15	N S	7		---	±0.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
16	N S	8		---	±0.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
17	N S	10		---	±0.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
18	N S	12		---	±0.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
19	N S	15		---	±0.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
20	N S	25		---	±0.8	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
21	N S	35		---	±0.9	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
22	N S	40		---	±1.0	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
23	N S	50		---	±1.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
24	N S	60		---	±1.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---

1/ VSWR value is for both ends.

2/ Peak power for a duty cycle of 5×10^{-4}, maximum pulse duration of 5 microseconds.

3/ Power input is derated linearly to 0.5 watts at +125°C.

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Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - MI
Navy - AS, MC, SH
Air Force - 99

Preparing activity:
DLA - CC

(Project 5985-1229-03)

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